SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares energy consumption and operational costs between baseline and all applied renovation scenarios from CSV results. Embodied carbon is annualized using an analysis period of 30 years, while operational carbon is reported over 1 year. The construction cost data is from 2026 RSMeans Database. The embodied carbon is calculated using data from EC3 Environmental Product Declaration (EPD) Database. Since RSMeans Database is proprietary, users can adopt customized cost dataset when needed.

Building Information

- Weather File: USA_TX_Amarillo-Husband.Amarillo.Intl.AP.723630_TMYx.epw
- Building Name: SmallOffice
- Building Location: Amarillo, TX
- Total Floor Area: 511 m²

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	Baseline (no envelope renovation)
Scenario 1	wall + roof + window + door	**Wall upgrade:** Exterior walls were upgraded to R-13.0 with Fiberglass Batts insulation. **Roof upgrade:** The roof was upgraded to R-24.4 using Blown Fiberglass insulation. **Window upgrade:** • Upgraded to 3-pane glazing • Weatherstripping: silicone adhesive smoke gasket

		• Glazing film: low-e film • Window frame: wood window frame • Caulking: acrylic • Infiltration reduction: 30% **Door upgrade:** • Door type: wooden door • Bottom seal: automatic door bottom • Top/side seals: jamb weatherstrip • Infiltration reduction: 30%

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Total Site Energy (GJ)	195.48	189.95	**5.53**	**2.83%**

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) ($/yr) Comparison

Baseline $9,667

Scenario 1 $9,201

Baseline Scenario 1

Max saving: **$-466 (-4.82%)**

Operational Carbon Saving (Scenario - Baseline) (kg CO_2e/yr) Comparison

Baseline 18,943

Scenario 1 18,423

Baseline Scenario 1

Max saving: **-521 kg CO_2e (-2.75%)**

Min Retrofit Construction Cost

$9,000

Scenario 1

Min Retrofit Embodied Carbon

0.00 kg CO_2e

Embodied carbon intensity (ECI): 0.00 kg CO_2e/m^2

Scenario 1

Lowest Retrofit Cost Payback Period

19 years

Lowest Retrofit Carbon Payback Period

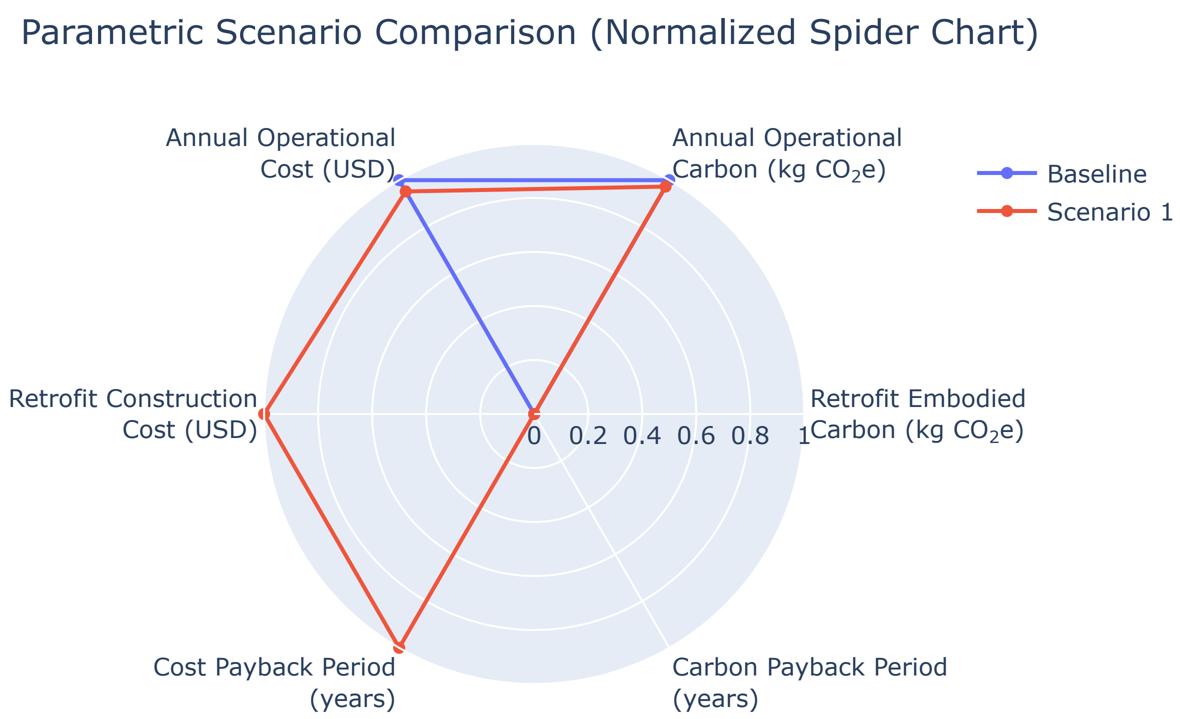
N/A

Scenario 1

No valid carbon payback scenario

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) ($/yr)).

Scenario 1 19 yrs

Carbon Payback Period

Payback = retrofit embodied carbon / abs(Operational Carbon Saving (Scenario - Baseline) (kg CO_2e/yr)).

Scenario 1 N/A

Material List

Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material	Volume (m^3)	Area (m^2)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m^3)
Scenario 1	Wall insulation	Fiberglass Batts	N/A	282	N/A	N/A	N/A	N/A
Scenario 1	Roof insulation	Blown Fiberglass	N/A	599	N/A	N/A	N/A	N/A
Scenario 1	Window weatherstrip	silicone adhesive smoke gasket	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 1	Window caulking	acrylic	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 1	Door bottom seal	automatic door bottom	N/A	N/A	N/A	N/A	N/A	N/A
Scenario 1	Door top/side seal	jamb weatherstrip	N/A	N/A	N/A	N/A	N/A	N/A

Result Summary

Scenario	Retrofit Embodied Carbon (kg CO_2e)	Retrofit Construction Cost (USD)	Annual Operational Carbon (kg CO_2e)	Annual Operational Cost (USD)
Baseline	0.00	$0.00	18,943	$9,667
Scenario 1	0.00	$9,000	18,423	$9,201

Generated: April 23, 2026 | Report Type: Retrofit Impact Analysis | Run: run_test_004